

2025 22 $\Rightarrow$ 40% 2

Lec_23

Recursive Tree method

3.) $T(n) = T(n/5) + T(4n/5) + n$

$$\Rightarrow \underline{n + \gamma + \gamma}$$

$$T(n) = T(n/5) + T(4n/5) + n$$

$$T(n/5) = T(n/5^2) + T(4n/5^2) + \frac{n}{5}$$

$$\tau(4n/5) = \tau(4n/5_2) + (\frac{4n}{5_2}) + \frac{4n}{5}$$

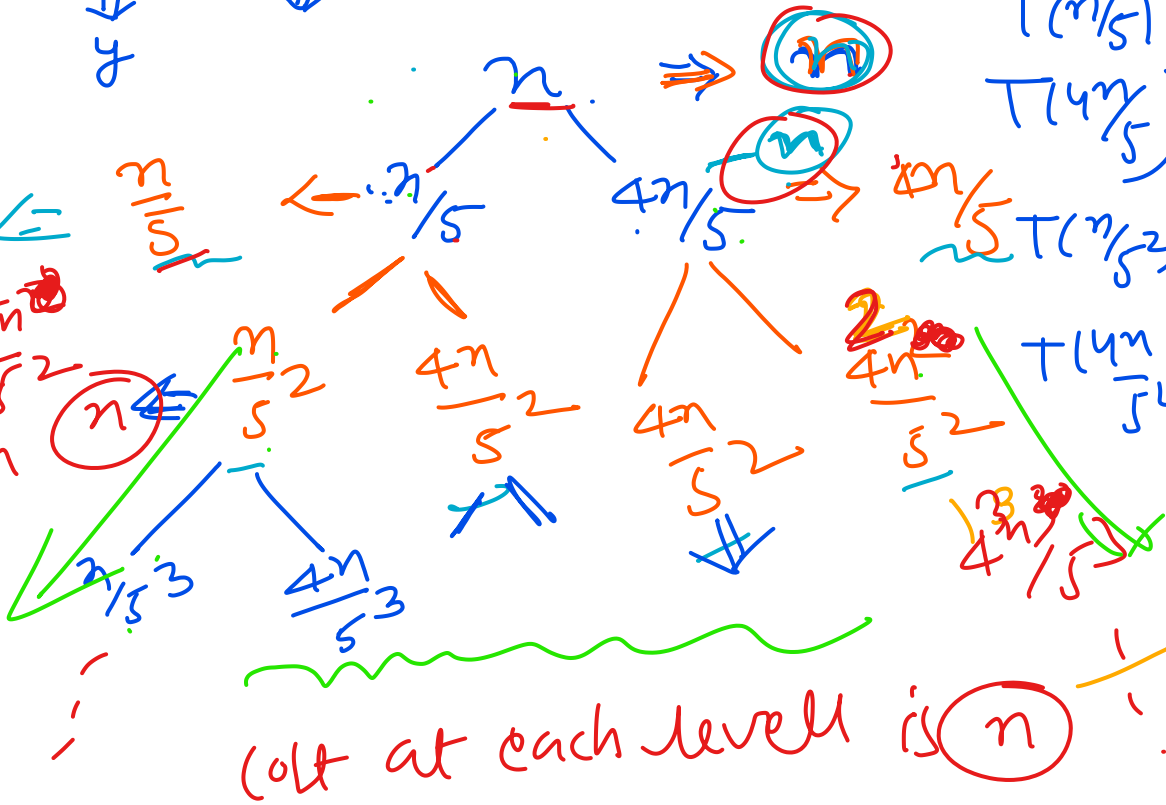
$$5T(n/5) = T(n/3) + T(n/3) + T(n/3) + n$$

$$T\left(\frac{4n}{5^L}\right) = \frac{4n^2}{5^3} + \left(\frac{4n}{5}\right) + \frac{4n}{5^2}$$

$$\frac{n}{\sqrt{2}} + \frac{4n}{\sqrt{2}} + \frac{4n}{\sqrt{2}} + \frac{4n^2}{\sqrt{2}}$$

$$\frac{n+4n+4n+16n}{5^2}$$

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cost at each level is n .

$$k_1 = \log_5 n$$

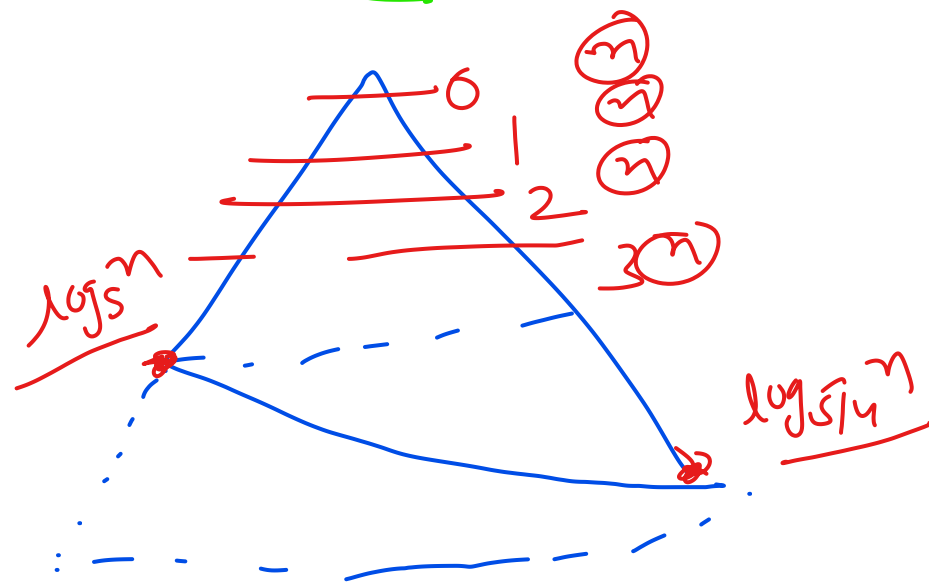
$$n/s^{k_1}$$

$$\frac{n}{s^{k_1}} = 1$$

$$n = s^{k_1}$$

$$\log_5 n = k_1$$

cost at each level is $\underline{\underline{n}}$



$$\begin{aligned} & n/s^2 \rightarrow n/s \rightarrow n \rightarrow 4n/5 \\ & 4^2 n / 5^2 \rightarrow 4^3 n / 5^3 \rightarrow 4^4 n / 5^4 \end{aligned}$$

$$\begin{aligned} & n/s^k \\ & n/s^k \end{aligned}$$

$$\frac{k_2}{\Delta n} = \left(\frac{4}{5}\right)^{k_2}$$

$$\left(\frac{4}{5}\right)^{k_2} = 1$$

$$n = \left(\frac{5}{4}\right)^{k_2}$$

$$\log_{5/4} n = \log_{5/4} (s^{k_4})$$

$$\Rightarrow k_2 = \log_{5/4} n$$

worst case:-

W.C $\Rightarrow n+n+n \dots$

$\log_5 n$ times

$$TC \leq \Rightarrow \frac{n \times \log_5 n}{5}$$

$$\frac{n+n+n+n+n}{5n}$$

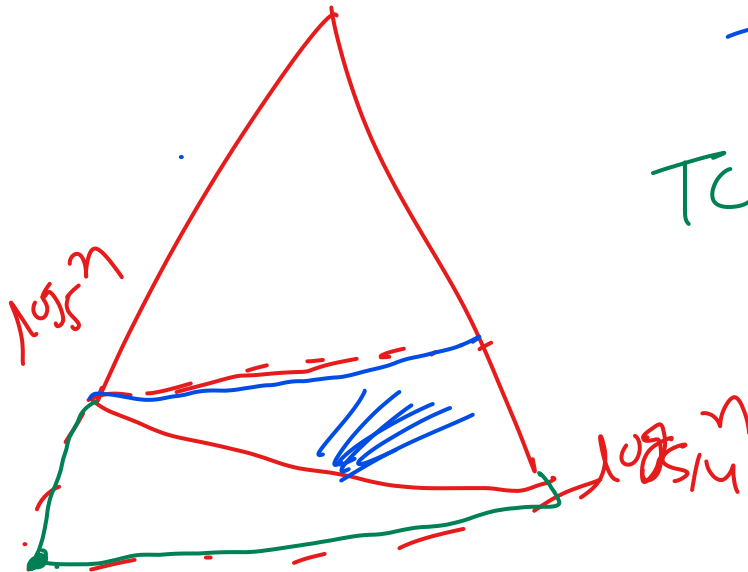
$$TC \Rightarrow O(n \cdot \log_5 n)$$

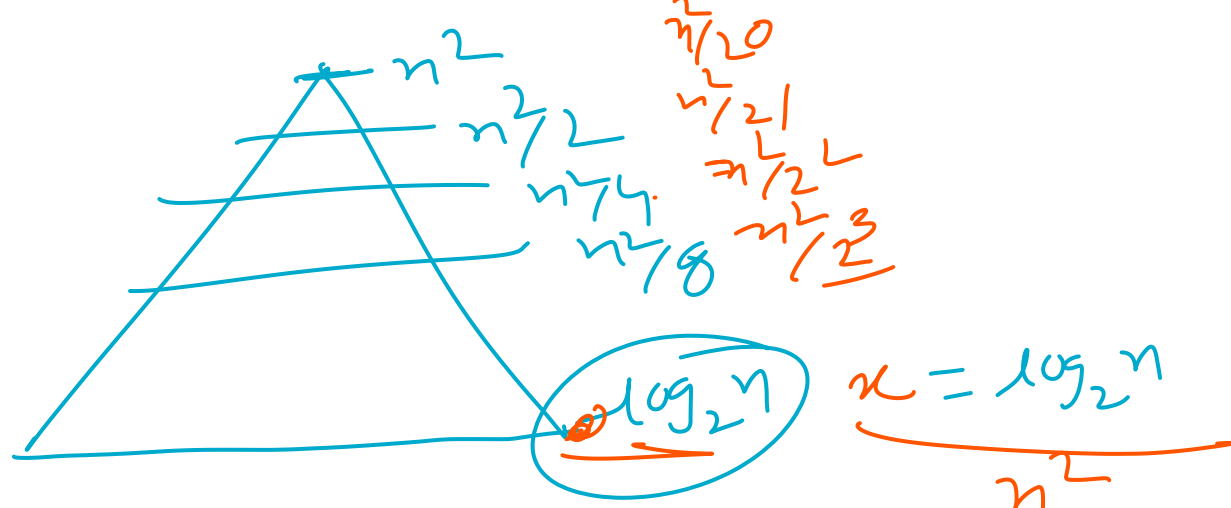
$$T(n) \geq n+n+n \dots \log_5 n \text{ times}$$

$$T(1) \geq 2 \times \log_5 n$$

$$T(n) = \Omega(n \log_5 n)$$

$$T(n) \checkmark$$





$$T(1) = n^2 + \frac{n^2}{2} + \frac{n^2}{4} + \frac{n^2}{8} - \dots - \frac{n^2}{2^k}$$

$$T(n) = n^2 \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} - \dots - \frac{1}{2^k} \right]$$

$$= n^2 \left[\frac{a(r^{n+1} - 1)}{r^{n+1} - 1} \right] \quad \text{decreasing a.p}$$

$$= n^2 \left[1 \left(\frac{\frac{1}{2}^{n+1} - 1}{\frac{1}{2} - 1} \right) \right]$$

$$= n^2 \left[\frac{2^{n+1} - 1}{2^{n+1}} \right]$$

$$= \frac{n^2}{2}$$

$$T(n) = \Theta(n^2)$$

$$n^2 \times O(1) = n^2$$

$$\begin{aligned} T(n) &\geq n^2 \\ T(n) &\leq n^2 \end{aligned}$$

Thanks

